Formal companion to Paper IV

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This document describes the frozen formalization supporting *Clique partitions with rooted simplicial defect: quantitative stability and sharp eventual bounds*. It accompanies the editorial revision of the manuscript; the mathematical statements and the Lean source freeze are unchanged. Section, equation and theorem numbers refer to the main manuscript unless otherwise specified. References [1–23] retain its bibliography.

1. Source identity and verification scope

Item	Frozen value
Lean and Mathlib	v4.28.0
Source cut	piv-v12-fb459343d234
Main source modules	607
Source/configuration/documentation manifest entries	615
Selected audit targets	19
Literal exports through `import PaperIV`	224
Recorded axiom outputs	461; not a count of distinct theorems

The source manifest SHA-256 is fb459343d234f968d7d32eff1491ea8a09aa2e135b313a80623012e7449042f5. The source-only archive SHA-256 is cb2741454380736ffe01b59933057b5079fa9f865d5bbdd3f67534bf808a62e6. These identify the original formal source package, not the new editorial PDFs.

The frozen package and recorded build evidence are available in the series repository. The series concept DOI is 10.5281/zenodo.21273143. The preceding manuscript edition and its deposit remain available; the source freeze is the same.

The identified Lean supplement for this version uses Lean and Mathlib v4.28.0. The final declarations do not take rounding, existence of an optimum or the near constructor as arguments: they invoke them as proved theorems in their dependencies. The permitted foundational axioms are distinguished from any additional mathematical hypothesis.

The recorded reconstruction covers all 607 modules: 421 objects built in the first segment were verified and reused after a restart, and the resumed segment compiled 186 modules. All 19 selected targets were freshly executed with exit code zero. Their 461 axiom outputs use only propext, Classical.choice and Quot.sound, or subsets. The records distinguish source identity, compilation, export availability and transitive axiom checks. None of these counts is a claim that every manuscript sentence is a separately exported theorem.

The consolidated external report for v1.22-r4 records PASS for E0–E8 within its documented scope. That report and the build logs remain evidence for the identified frozen source. They are not a fresh external review of v1.24. The present editorial review checks relocation, statement preservation and document rendering; no Lean build or new mathematical verification is claimed.

2. Mathematical statements and exported declarations

The common prefix PaperIV is omitted where indicated. The following map preserves the original theorem labels; moving it out of the main manuscript has not changed the formal interface.

Result	Lean declaration
Theorem A, additive form	Erdos81AllOrders.erdos81_all_orders_additive
Theorem B, upper bound	Erdos81Unconditional.erdos81_cliquePartition
Theorem B, maximum	PaperTheorems.erdos81_max_eq
Theorem 3.1	RC01Final.rc01_uniformRoundingTarget
Corollary 3.5	FarRegimeAllGraphs.farRegime_cliquePartition _allGraphs
Dichotomy (1.5)	HybridDichotomy.chordal_far_or_nearStructure
Theorem 5.0, strengthened window	NearH1LocalConstructor.exists_near_partition _paid_by_root_sharp
Theorem 5.0, previous interface	NearH1LocalConstructor.exists_near_partition _paid_by_root
Corollary 5.4, size and density	NearCriticalDichotomy.chordal_far_or_criticalR oot, chordal_far_or_edge_density
Optimal quadratic coefficient	SharpConstantOptimality.erdos81_quadratic_co nstant_optimal, erdos81_quadratic_constant_isLeast
Theorem 6.1, strengthened form	IntegralStability.chordal_linear_stability_sixteen
Theorem 6.1, earlier interface	IntegralStability.chordal_linear_stability
Corollary 6.1a, piece count	RootPartitionStability.chordal_partition_stabili ty
Corollary 6.1a, joint real piece and edge bounds	SublinearResearch.chordal_joint_stability_real in PublicationStability
Identity (6.7b)	RootPartitionStability.sum_rootPieceDefect_eq
Corollary 6.2	ExtremalClassification.chordal_extremal_classifi cation
Proposition D.3	SplitMixedGap.mixed_gap_zero together with the bounded construction in SplitCompleteSharpValue

For the extension table below, the prefix is PaperIV.SublinearResearch, except for the named E32 entry.

Statement	Declaration
Real same-root chordal bounds, including edge mass	chordal_joint_stability_real
Theorem C' and (6.12)–(6.13)	fixed_defect_joint_stability_real
Explicit parameters and classification	FixedExplicit.fixed_defect_stability_explicit
Optimal-size edit bound (6.17)	fixed_defect_exact_extremal_edit_real
Unrestricted equality classification	E32.cp_classification_of_theoremCPrimeC, with the proved classification above and the historical induced-removal assembly [23] (see companion §3)

Statement	Declaration
Finite bound (6.22)	certified_shifted_fractional_bound_all_orders
Finite bound (6.23)	certified_refined_fractional_bound_all_orders
Uniform integral versions	uniform_shifted_partition_bound; uniform_refined_partition_bound
Arbitrary-order sequence approximation	sequence_arbitrary_orders_partition_bound
One-root sequence stability	sequence_arbitrary_orders_stability

The finite same-root interfaces take real deficits. The sequence interface records rational error and excess parameters. That difference is retained when comparing the manuscript with elaborated types.

Literal definitions. The following excerpts reproduce definitions and theorem headers, omitting proofs but not hypotheses. The first excerpt is in the namespace SimpleGraph; the next three are in PaperIV.FarRounding, with finite V, decidable equality and decidable adjacency for G. The last definition is in PaperIV. The source locations are PaperIV/ChordalStructure.lean (chordality), PaperIV/FarRounding.lean (pairs and partitions), and PaperIV/TargetEnvelope .lean (target size), within the frozen package linked in §1 of this companion.

```
def IsChordal (G : SimpleGraph V) : Prop :=
  ∀ ⦃v : V⦄ (c : G.Walk v v), c.IsCycle → 4 ≤ c.length →
    ∃ x y : V, x ∈ c.support ∧ y ∈ c.support ∧ G.Adj x y ∧ s(x, y) ∉ c.edges
def pairs (K : Finset V) : Finset (Sym2 V) := K.sym2.filter fun e => ¬ e.IsDiag

structure CliquePartition where
  pieces : Finset (Finset V)
  isClique : ∀ K ∈ pieces, ∀ a ∈ K, ∀ b ∈ K, a ≠ b → G.Adj a b
  two_le_card : ∀ K ∈ pieces, 2 ≤ K.card
  edgeDisjoint : ∀ K ∈ pieces, ∀ L ∈ pieces, K ≠ L → Disjoint (pairs K) (pairs L)
  covers : pieces.biUnion pairs = G.edgeFinset

def CliquePartition.size (Q : CliquePartition G) : ℕ := Q.pieces.card

def CliquePartition.OrderAtMost (Q : CliquePartition G) (r : ℕ) : Prop :=
  ∀ K ∈ Q.pieces, K.card ≤ r
def targetSize (n : ℕ) : ℕ := n * (n + 1) / 6
```

Natural-number division implements the floor in $M(n)$. The expository symbol c_4 denotes the minimum over these partitions with `OrderAtMost 4`; the exported statement does not assume a minimum function, but directly supplies a witness partition. Chordality in FarRounding is a literal alias of the preceding cycle-based definition.

Exported theorems. In PaperIV.Erdos81AllOrders and PaperIV.Erdos81Unconditional, respectively, the exact headers are:

```
theorem erdos81_all_orders_additive :
    ∃ b : ℕ, ∀ (n : ℕ) (G : SimpleGraph (Fin n)) [DecidableRel G.Adj],
      PaperIV.FarRounding.IsChordal G →
        ∃ Q : CliquePartition G, Q.OrderAtMost 4 ∧ Q.size ≤ PaperIV.targetSize n + b
theorem erdos81_cliquePartition :
    ∃ N : ℕ, ∀ n : ℕ, N ≤ n → ∀ (G : SimpleGraph (Fin n)) [DecidableRel G.Adj],
      PaperIV.FarRounding.IsChordal G →
        ∃ Q : CliquePartition G, Q.OrderAtMost 4 ∧ Q.size ≤ targetSize n
```

These headers allow the quantifier order and exact coverage to be checked without inferring them from theorem names. The recorded audit contains no sorryAx among the axioms of the examined declarations. The supplementary report distinguishes that transitive check from the textual search for native_decide in project sources and from a complete rebuild of dependencies.

3. Dependencies, adapters and attribution

Result in the text	Reference module
Inputs and construction	
Attained rational optimum	PaperI.FiniteLPDuality, CertifiedOptimumExistence
Uniform mixed rounding, Theorem 3.1	RC01Final, MixedRoundingAdapter; Nibble.* infrastructure from Paper III
Corollary 3.5, far regime for all graphs	FarRegimeAllGraphs.farRegime_cliquePartition _allGraphs; explicit version in E17.ExplicitFarAssembly
Complete improved cleaning and explicit threshold	E17.Cleanup, E17.FarBudget, E17.ExplicitFarRounding, E17.ExplicitFarAssembly; constants from E18 and E19
Fractional nibble selection, Paper III	Nibble.FracNibbleLE, Nibble.FracNibbleRepaired; adapters PaperIIISlackNibbleAdapter, PaperIIINibbleAdapter
Localization, root and critical window	NearH1Localization, NearRootWindow, RootRegularizationBridge
Dichotomy and absorbable interface	HybridDichotomy, SeparationAbsorptionRoute
Accounting and near partition	RD09PhysicalLedger, NearH1FinalAssembly, NearH1LocalConstructor
Extremal conclusions	
Sharp bound and eventual maximum	Erdos81Unconditional, PaperTheorems
Bounds for all orders	Erdos81AllOrders; explicit bound in ExplicitThreshold
Fixed rooted defect and maximum, Theorem C	DefectExplicitPublication; explicit assembly E34.L11Main; lower witness DefectSharpPublication
Necessity of order four, Proposition D.2	DefectSharpPublication.order_three_insufficient, E11K10
General budget and passage to cp	LossBudget, SplitMixedGap
Graph stability, partition stability and extremal classification	IntegralStability, RootPartitionStability, ExtremalClassification
Complete-split value and cores	SplitCompleteSharpValue, SplitCompleteRigidity
Partition defect and optimal coefficients	SplitCompleteDefect, LinearCoefficient, SharpConstantOptimality
Edge density in the critical branch	NearEdgeDensity, NearCriticalDichotomy
Equality of mixed optima, Appendix D.2	SplitMixedGap, SplitCompleteSharpValue
Complementary tools	
Budgeted compatible absorption	SpreadAbsorptionCompatibility
Balanced reserve, Appendix B.2	BalancedReserve, Nibble.BeckFiala from Paper III
Four-host completion, Appendix B.3	FourHostClosure; Vizing and MultiHostTriangleLift
Exact reserve and threshold sensitivity	ReserveIdentity, NearThresholdSensitivity
Reusable chordal structure	CliqueTree

Module map. The prefixes PaperI, Nibble and MixedRounding indicate, respectively, the rational development associated with Paper I, the nibble infrastructure of Paper III and the neutral mixed-model library. The prefixes E17, E18, E19 and A4S1 identify the increment modules integrated into the project; other unprefixed names belong to PaperIV; MixedRoundin gAdapter adapts that namespace to the MixedRounding library. RC01 incorporates and adapts tools from Paper III, but its two-quota mixed assembly is the one in Section 3 of the present work. The table relates results to sources; the supplement retains the exact dependency list.

Chordal structure and Dirac's theorem come from the Paper II contribution, ported into ChordalStructure. Szemerédi's regularity lemma is reused from Mathlib; the construction and adapters for clean profiles and mixed selection belong to Paper IV. This provenance is distinct from a file's current location under PaperIV.

The configuration declares Mathlib as a third-party package and a selection of Paper III Nibble modules as a local dependency of the series. ConeAudit traverses the types and proof terms of the selected declarations and checks that their transitive dependencies contain no names under the root Erdos81, used by [5]. Our modules PaperIV.Erdos81Unconditional and PaperIV.Erdos81AllOrders have namespace root PaperIV; they are not the excluded root Erdos81 used by [5]. The absence of these names is a reproducible audit result, not an inference from the import list. Its scope is precise: it certifies that exclusion in the examined snapshot; it would not by itself detect relocated and renamed code or determine the historical provenance of each argument. Source attribution is documented separately. The same namespace check cannot yield a verdict on [15]. The extension uses the signed-star method of [15, Lemma 3.1] and the removal method of [22]; proving their implementations in this supplement does not make those methods new.

The public interface for Theorem C in this revision is DefectExplicitPublication: it uses E34.theoremC_fully_explicit_final and the unrestricted lower witness already proved in DefectSharpPublication. The older existential assembly is not the proof selected for Theorem C. Its localization uses induced removal of Alon–Shapira [23]; the selected explicit assembly instead uses the polynomial chordal-removal development of [22]. The additional FDCheck.AS Check traverses proof terms and types to test exclusion of the named removal results in these selected declarations. This is not exclusion of the AlonShapira namespace: shared definitions of chordality, edit distance and induced copies remain in use.

There is a specific exception to this dependency comparison. The unrestricted equality classification in Corollary 6.3, (6.18), is exported through E32.cp_classification_of_theoremCPrimeC; that declaration and E32.ref15Theorem11_of_theoremCPrimeC still use the historical assembly through E32.theoremC_at. Their proofs therefore include the induced-removal chain of [23]. All those lemmas are proved in the frozen sources with standard axioms. The exclusion check for the explicit Theorem C and the selected stability declarations must not be extended to these classification wrappers.

3.1. Resources and the two gain conventions

The representation refinement in the original proof map is verified in MixedRoundingAdapter. The adapter identifies a piece's edges and the admissible-copy predicate, transports fractional packing capacities, and proves equality of values via value_toFarFrac. For an integral packing, gain_toFarPacking preserves gain. Thus Theorem 3.1 can be used through this neutral presentation without changing budget (1.3). This is not a new proof of rounding.

Two functions named gainOf must be distinguished by their domain of use. Model.gainOf is two at cardinality three, five at cardinality four, and zero at all other cardinalities. FarRounding.gainOf is $\binom{|K|}{2} \dot{-} 1$, where $\dot{-}$ denotes truncated subtraction in the naturals. They agree at cardinalities zero through four, but cease to agree from five onwards. Mixed items have cardinality three or four, so the transport is legitimate. For unrestricted partitions, such as those in Section 6.4, one uses edge accounting or FarRounding.gainOf; it is never replaced by the bounded gain function from Model. Since partition pieces have at least two vertices, truncated subtraction there agrees with ordinary subtraction.

3.2. Construction and retained accounts

The bounded-rank nibble is reused through PaperIIISlackNibbleAdapter.boundedRankNibble At. The two-quota construction is in MarkedQuotaSlackGate.slackMarkedQuotaNibbleAt_pr oved; the pairing of triangular supports and its physical realization are in MarkedQuotaPairing and JointTwoQuotaPhysical. The joint codegree scale corresponds to RC01CleanedGate.clea nedPacking_joint_codegree_le_of_served_patterns. Profile mass is at most t^2, not one: this is the physical normalization of RC01PatternMassScale.

For the mixed path, VertexCopyMonotone.two_mul_F4_le_add proves the two-direction inequality. VertexCopyGate selects the step and its finite potential; GatedTerminalSplit.exi sts_split_terminal_symmetrizationPath supplies the complete sequence. Local accounting is constructed in NearH1WindowAccounts before being imported into NearH1Localization. This dependency direction matches the logical order explained in Lemma 4.2.

Clique extraction and calibration are in ChordalCoreMissing and NearH1CalibratedRoot. Regularization, RegularizationBounds and RegularizedRootGoal prove the root properties used by RootRegularizationBridge. Finally, NearH1PhaseI, RD09FactorCandidateMoments and RD09FactorCandidateAverage produce the two accounts; NearH1FinalAssembly combines them using the same first assignment and its occupied links. The second phase is not applied to an independent copy of the resources.

Theorem 5.0 has a strengthened form, NearH1LocalConstructor.exists_near_partition_pai d_by_root_sharp, which adds $|3|R| - n| \le n/50$ to the physical constructor; the previous interface exists_near_partition_paid_by_root remains as a corollary. The strengthened reserve in (5.15a) is retained for the same witness in NearStructureWitness.accounts_co unt_le_improved; IntegralStability.chordal_linear_stability_sixteen and RootPartitio nStability.chordal_partition_stability use it without recombining roots or packings. The cyclic selection in phase I is in RD09PaddedL1Mass; the final evaluation of phase II, in H1ImprovedConstants.l10_of_raw_rd09L2. The comparator regimes in (5.1a) are checked in SplitComparatorResidual.residual_sq_le_of_near_split_universal. Reserve (D.8) is ReserveIdentity.targetSize_eq_baseline_add_reserve; the improved contraction corresponds to NearThresholdSensitivity.descent_threshold_maximal_barrier and near_threshold _binding. The critical-geometry results are in NearRootWindow, NearEdgeDensity and NearCriticalDichotomy.

The extension results have separate public entry points. Theorem C and its maximum are PaperIV.DefectExplicitPublication.rooted_defect_eventual and rooted_defect_maximum; the historical DefectSharpPublication namespace retains the lower witness and order_three_ insufficient for Proposition D.2. E35.theoremC_tower and its uniform variants give Corollary 6.6. The complete account (3.11) is E17Bridge.improved_far_loss_explicit. Corollary 3.5 with an explicit threshold is E17Bridge.ExplicitAssembly.farRegime_allGraphs_explicit. Finally, PaperIV.ExplicitThreshold.erdos81_sharp_explicit and erdos81_all_orders_bounded_add itive give the global result with the tower of §6.3. These names identify declarations, not new hypotheses.

3.3. Scope relative to other formal developments

The formal scope differs from that documented by [5] at the cited commit. Its lean/FORMALI ZATION_STATUS.md file declares verification conditional on ExternalInputs.Inputs: Vizing, Häggkvist–Janssen and a packing transfer that includes certificates of rational optimality. These inputs are explicit parameters, not hidden axioms. Here the inputs used by our chain, including Vizing, are discharged; Häggkvist–Janssen is not used. The comparison concerns what each Lean development checks: it does not assert that the mathematical proof of [5] is conditional

on conjectures.

Reusable contributions have their own references. Sum-zero triangle packing corresponds to Lean Pool PR #348 [12], https://github.com/Vilin97/lean-pool/pull/348, merged at commit 540d8e3. Minimum-degree matchings and their weighted selection correspond to PR #420 [9], https://github.com/Vilin97/lean-pool/pull/420, merged at d1de6d2. The series sources retain the antecedents and adaptations of Vizing and Beck–Fiala. Attribution of formalizations is distinguished from authorship of the classical theorems.

Lean [10] and Mathlib [11] provide the verification environment. Code auditing does not replace human review of definitions or of the correspondence between a formal statement and its expression in the manuscript. Experiments guided the search and detected false assertions; they are not premises of universality.

The formal constructor used here invokes neither the results of [15] nor the historical modules of the replaced construction. The list bound and first-phase capacity window have the same form as conditions in [15, Lemma 4.1 and (4.4)]. Here they are justified by dyadic hosting, averaging over cyclic shifts, and balanced absorption. This comparison identifies the mechanisms and their shared conditions; it claims neither priority for those conditions nor historical independence on the basis of the dependency check in §3 of this companion.

4. Precision notes for the manuscript

The following distinctions survive the removal of implementation names from the exposition. They prevent a mathematical statement from being identified with a stronger or differently quantified Lean type.

- The construction in §6.1 uses triangles and edges; the audited public interface supplies a partition of order at most four and an unrestricted lower bound.
- The quadratic and linear optimality declarations use rational coefficients. The manuscript explains the passage to real coefficients by rational approximation.
- recover_clique_from_edit takes rational $u \geq 0$; the written proof also applies to real $u \geq 0$.
- Proposition G.1 uses a rational precision parameter in the Lean declaration and real exponential comparisons.
- The width formulation in Appendix G.1 uses clique trees with bags of bounded cardinality; it does not rely on a general Mathlib treewidth definition.
- No chordal restriction of the cleanup-to-rounding implication is formalized by Appendix G.2. The universal obstruction is not a lower bound for every chordal rounding method.

4.1. An existing numerical premise made explicit

The old prose in F.5 referred to thm2_ineq by name. The new prose prints its existing conclusion, with $m = m_{11}(\varepsilon, K)$, $B = B_{11}(\varepsilon, K)$ and δ_{11} as in (F.11). The source is E34/L11P arams.lean, lines 289–292. Its assumptions are $0 < \varepsilon < 1$ and $1 \leq K$; the preceding paragraph of F.5 supplies the stronger $\varepsilon < 1/2$ and $K \geq 1$. This is an expansion of a cited premise, not a new estimate or proof.

4.2. Source map for relocated inline references

Manuscript location	Formal declarations or modules
1.1. The common condition: a loss budget	LossBudget.budget_iff
1.2. Structural dichotomy and resolution mechanisms	NearStructureWitness
5.3. The three accounts that pay for the construction	isSpokeCompatible_surviving_canonicalBad, RD09SpokeCompatibility.card_union_phases
6.2. The eventual maximum	SplitCompleteRigidity.optimal_cores
6.2. The eventual maximum	PaperTheorems.splitGraph_isChordal
6.3. All orders and optimality of the linear term	SharpConstantOptimality.erdos81_quadratic_constant_optimal, erdos81_quadratic_constant_isLeast
6.3. All orders and optimality of the linear term	LinearCoefficient.linear_coefficient_optimal
6.3. All orders and optimality of the linear term	LossBudget.erdos81_cp_form_all_orders, erdos81_cp_form
6.6. Fixed-defect stability on one root	NormalizedDeficit, ResidualExcess, NormalizedCliqueCore, RootPartition, NormalizedStability
6.7. Uniform bounds and sublinear defect	E35.Fexp_le_tower
B.1. Budgeted compatible absorption	SpreadAbsorptionCompatibility.exists_budgeted_compatible_spread_absorber
B.2. Balanced reserves	BalancedReserve.exists_balanced_reserve
B.3. Completion with four hosts	FourHostClosure.exists_fourHost_packing
C.2. What is discarded and why the threshold is uniform	RC01CleanFiber, RC01DeviationCleanup
C.3. An explicit schedule and its propagation	SzemerediRegularity.initialBound
C.3. An explicit schedule and its propagation	E18.NibbleSchedule, NibbleOracle, NibbleChain
C.3. An explicit schedule and its propagation	E19.ScheduleBounds, E19.ChainBounds, E19.GateBounds
C.3. An explicit schedule and its propagation	E18.Numeric, E19.Main
C.3. An explicit schedule and its propagation	E35.NfarE_le_tower_poly
D.1. The complete graph at every order	ThreeRegime.CompleteStateAllOrders
D.1. The complete graph at every order	HalvingFrames
D.1. The complete graph at every order	exists_packing, complete_state_closes
D.2. Equality of mixed optima in complete-split graphs	SplitMixedGap.mixed_gap_zero, SplitCompleteSharpValue
E.1. Constructor and accounting	IndepAllBounds, IndepAllParams
E.1. Constructor and accounting	IndepAllPeel, IndepAllObstr
E.1. Constructor and accounting	IndepAllBudget
E.1. Constructor and accounting	IndepAllMain, ConstructorBudget
E.2. From the terminal to the eventual theorem	E34.theoremC_fully_explicit_final
E.4. Retained accounts for same-root stability	StrictParams
E.4. Retained accounts for same-root stability	StrictBudget
E.4. Retained accounts for same-root stability	ResidualExcess
E.4. Retained accounts for same-root stability	AllInput.no_core_pairs
E.4. Retained accounts for same-root stability	CoreCliqueAlternative
E.4. Retained accounts for same-root stability	TemplateDistance, RootPartition, CliqueShiftArithmetic
F.1. Explicit fixed-defect parameters	ExplicitFixedStability, E33.Assembly, E33.LocExplicit, E34.L11Main
F.1. Explicit fixed-defect parameters	fixed_defect_stability_explicit
F.2. Fractional star accounting and small orders	RefinedFractionalBound, UniformFractionalBound, SmallOrderFractional
F.3. Uniform localization through a chordal edit	SublinearEdit
F.3. Uniform localization through a chordal edit	exists_clique_additive_defect, recover_clique_from_edit
F.3. Uniform localization through a chordal edit	SublinearStabilityTransfer
F.5. Auxiliary lemmas and adaptations of chordal removal	E34.indCopies_cycle_le

Manuscript location	Formal declarations or modules
F.5. Auxiliary lemmas and adaptations of chordal removal	E34.coBipartite_edit_le
F.5. Auxiliary lemmas and adaptations of chordal removal	E34.exists_large_clique
F.5. Auxiliary lemmas and adaptations of chordal removal	E34.lemma3
F.5. Auxiliary lemmas and adaptations of chordal removal	E34.transfer
F.5. Auxiliary lemmas and adaptations of chordal removal	E34.lemma11_V
F.5. Auxiliary lemmas and adaptations of chordal removal	properOn_of_pinned
F.5. Auxiliary lemmas and adaptations of chordal removal	thm2_ineq
F.5. Auxiliary lemmas and adaptations of chordal removal	claim5_count, block_ineq
F.5. Auxiliary lemmas and adaptations of chordal removal	l11_caseA, l11_caseB1, l11_caseB2, E34.L11Main
F.5. Auxiliary lemmas and adaptations of chordal removal	E35.theoremC_tower, theoremC_tower_uniform, theoremC_tower_uniform_sMax, NfarE_le_tower_poly, PaperIV.DefectExplicitPublication, FDCheck.ASCheck, PaperIV.OptimalTemplateObstruction.shifted_template_witness, shifted_template_sqrt_witness, no_linear_optimal_template_bound, optimal_family_nonempty
G.1. The all-order bound and packing gaps	BoundedCliqueGap
G.1. The all-order bound and packing gaps	BoundedCliqueGap.chordal_gap_linear_cliqueFree
G.1. The all-order bound and packing gaps	exists_cliqueTree_width_le_iff_cliqueFree
G.2. A quantitative obstruction to codegree cleanup	CleanupAtWith
G.2. A quantitative obstruction to codegree cleanup	threshold_gt_exp, threshold_gt_exp_seventy_two, FarExploration.CleanupRigidVerdict
G.2. A quantitative obstruction to codegree cleanup	threshold_superpolynomial
G.2. A quantitative obstruction to codegree cleanup	CodegreeCleanupAt, UniformRoundingTarget

5. Reproduction and separate annex

Run the following from the extracted, unchanged source archive with the toolchain pinned by lean-toolchain and dependencies pinned by lake-manifest.json. The command is documented for reproduction; it was not executed for this editorial revision.

```
python tools/audit_publication.py
```

It reads the 19 targets in FREEZE_SCOPE.json, runs them serially and keeps individual logs. Reimporting a compiled audit module is not equivalent to rerunning its commands. The recorded run is under 03_reproducibility/full_rebuild_v12_20260929_resume/, with provenance links to the first segment. ReleaseExportCheck uses only `import PaperIV`; the named audits inspect their own lists of types and proof terms.

Target
PaperIV
PaperIV.Audit
PaperIV.ConeAudit
PaperIV.PublicationIncrementAudit
A4S1.IndepAllAudit
ThreeRegime.AxiomCheck
FarExploration.AxiomCheck
PaperIV.CliqueTreeAudit
E17.ImprovedFarRoundingAudit
E17.ExplicitFarAudit
PaperIV.ExplicitThresholdAudit
ReleaseExportCheck
PaperIVEditorial.ConstructorBudgetAudit
ResearchAudit
FDCheck.FinalAudit
E35.Audit
FDCheck.ASCheck
E34.Audit
PaperIV.OptimalTemplateObstructionAudit

The literal source manifest is 03_reproducibility/build_piv-v12-fb459343d234/SOURCE_MANIFEST.json. The export log records 224 literal #check commands. The constructor audit covers PaperIV.Editorial.ConstructorBudget, which keeps both conclusions for one partition. The old extraction workspaces and superseded source cuts remain historical records in the series repository; they are not extra modules of the freeze described here.

BoundedCliqueGap is a separate source annex supporting Appendix G.1; it is not an input to the main extremal proof. Its source archive SHA-256 is 2847a422865e06880d457ea806aae5b9f749d22359eff325349c09c01cb4837d. Its 38 source modules and 50-module build closure belong to its distinct recorded build, not to the count of 607 above.

6. Written-proof review and limits

Scope of the written proof. The table below records seven derivations examined in the consolidated external review. Its assessment, ACCEPTABLE_SUMMARY for each row, retains the stated expository limit concerning the cited second-moment estimates in C.2. A successful build checks formal derivations, not completeness of exposition; the written-proof assessment remains distinct from compilation and from human peer review. The audit history and detailed assessments are retained with the supplementary evidence.

Location	Derivation to review, including the expanded accounts
§5.1, (5.4) and (5.7)	Regularization mass, degree, palette and width estimates
C.2, (3.8) and (3.10)	Discard counts and retention under the partition conventions
Lemma 3.2 and C.1–C.2	Fibre lower bounds from regular pairs and their normalization
C.3, Table C.1	Numerical domination of the nibble constants and tower iterations
E.1	Normalization bounds preceding the terminal construction
E.1, (E.2b) and (E.2d)	Terminal estimates and the three-case budget

Location	Derivation to review, including the expanded accounts
D.1	Bose/Skolem frame constructions underlying the residue table

Experimental checks of small orders are not used as premises. Their documentation, including completeness and certification requirements, belongs to the separate research record.

The clique-tree library retains counterexamples to formulations missing the appropriate hypotheses: a leaf alone is not equivalent to a unique simplicial vertex, and separator statements must account for repeated bags and degenerate cases. These results delimit the library's use; they add no assumptions to the main theorem.

The editorial ledger retains the exact source paragraphs and every replacement. Automated checks compare displayed equations, labels and result headings; hypotheses and proof dependencies are also reviewed in the source-to-manuscript comparison. These checks do not automatically certify semantic equivalence of all prose. The AI/tools disclosure retains its attribution and responsibility statements; the superseded draft-status sentence has been removed.

Use of artificial intelligence and computational tools

Claude, by Anthropic, and ChatGPT/Codex, by OpenAI, were used to explore and check arguments and to prepare the manuscript. Aristotle, by Harmonic, contributed to exploring approaches, searching for counterexamples, developing proofs, and formalizing and reviewing them in Lean; its role was not limited to translating finished proofs.

Certo [13], https://github.com/jtraverso/certo-math, and Jacobian [14], https://github.com/morluto/jacobian, were the computational tools used to explore instances, check identities, and assess candidates. Jacobian is also credited by the project behind [5]; its use here is acknowledged separately from the logical dependencies of our formal proof. Computations were reviewed within their stated scope; a finite search does not replace a universal proof.

The author remains responsible for the arguments, citations, code, and presentation. No AI system is listed as an author.

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